

Mizuho Quant Analytics Workshop

Getting Started

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MHI

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Global Quant Analytics

- MLIB is the Global Quant Analytics library at Mizuho
- Co-Developed by Mizuho Tokyo, London and New York

Features

- Yield Curves with CSA Support
- Swaps Pricing & Risk
- Bond Pricing: US Treasuries, German Bunds, Japanese JGBs et al
- IR Options: Swaptions, Caps, Floors et al
- Exotic Options: IR / FX Hybrids

- ① Getting Started
- ② Developer Tools
- ③ MLIB Ribbon
- ④ MLIB Functionality
- ⑤ Holiday Calendars
- ⑥ Daycount Conventions
- ⑦ Date Functions
- ⑧ Date Schedules
- ⑨ Utility Functions
- ⑩ Yield Curves, Forward Rates & Discount Factors
- ⑪ Swap Pricing & Risk
- ⑫ Swaps Pricer Tool

Excel Launcher

X:/Secure/DIR_MLIB_Analytics/MLIB_Excel_Addin

Features

- Function groups for navigation ease: meCurve, meProducts ...
- Multi Yield Curve construction & OIS Discounting with CSA Support
- OIS, IRS, Xccy and Asset Swap Pricing and Risk
- Every function has an Excel Example Template

Documentation

X:/Secure/DIR_MLIB_Analytics/Documentation

Examples

X:/Secure/DIR_MLIB_Analytics/Examples

Workshops

X:/Secure/DIR_MLIB_Analytics/Workshops

Automation Dashboard

MLIB Dojo

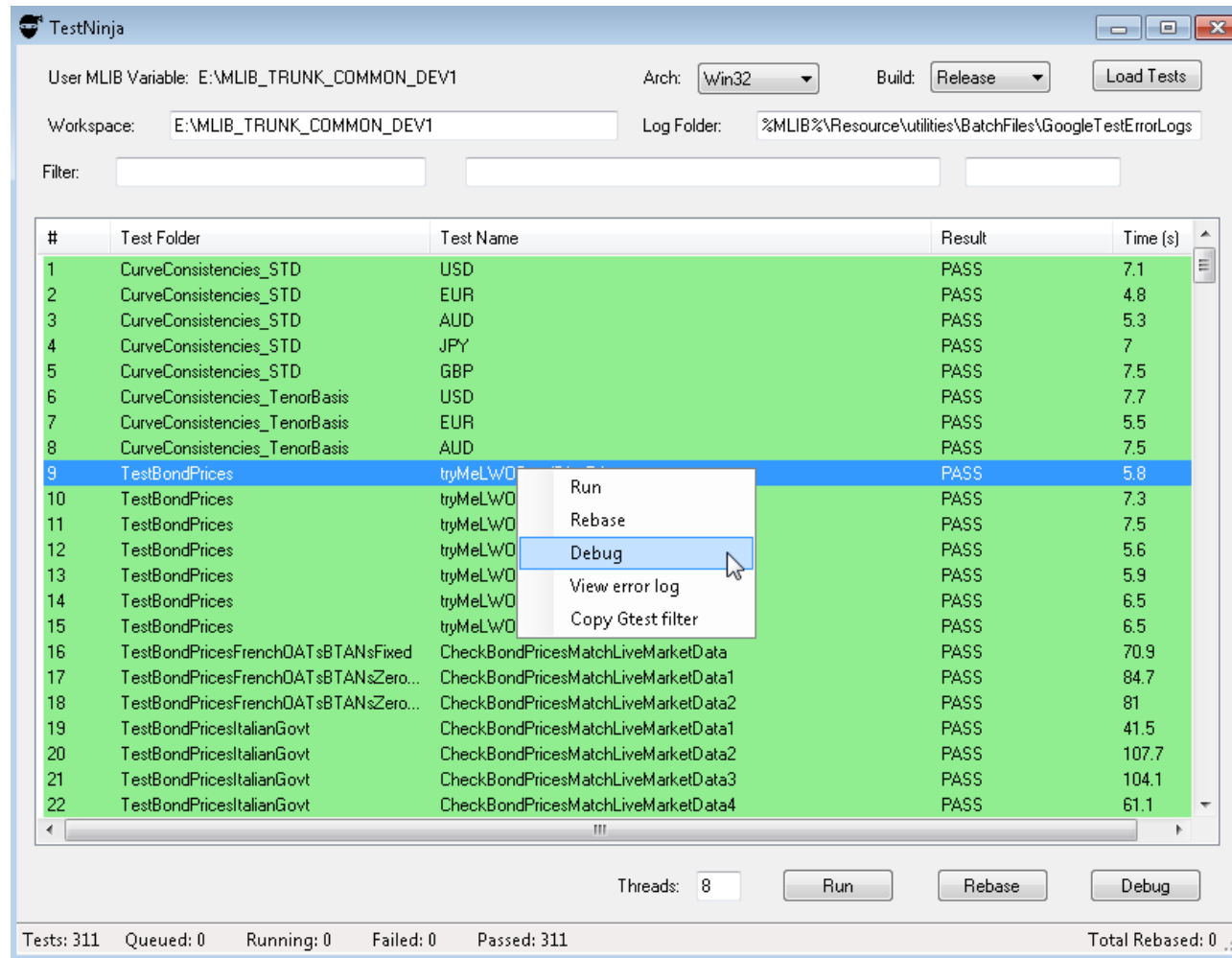
Read MLIB Variable Load Default Config Save Default Config Save My Local Config

User MLIB Variable: E:\MLIB_TRUNK_COMMON_DEV1

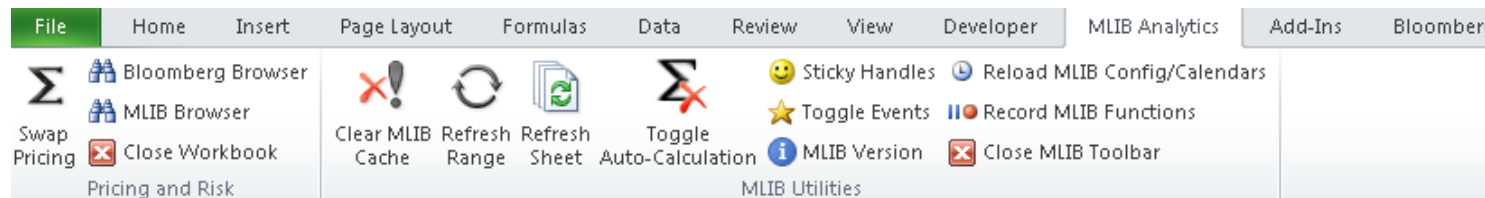
	Description	Batch File Path	実行 !
1	Set %MLIB% to DEV1	E:\MLIB_TRUNK_COMMON_DEV1\resource\utilities\BatchFiles\SetMibEnvironmentVariables\Set_MLI	Run 1
2	Set %MLIB% to DEV2	E:\MLIB_TRUNK_COMMON_DEV2\resource\utilities\BatchFiles\SetMibEnvironmentVariables\Set_MLI	Run 2
3	Perform SVN Update	%MLIB%\resource\utilities\BatchFiles\UpdateSVN\UpdateSVN.bat	Run 3
4	Build thenTest Debug x86	%MLIB%\resource\utilities\BatchFiles\BuildThenTest\BuildThenTestDebugx86.bat	Run 4
5	Build thenTest Release x86	%MLIB%\resource\utilities\BatchFiles\BuildThenTest\BuildThenTestReleasex86.bat	Run 5
6	Launch TestNinja	%MLIB%\resource\utilities\BatchFiles\TestNinja\TestNinja.exe	Run 6
7	Run Tests: Debug x86	%MLIB%\resource\utilities\BatchFiles\RunGtest\Run_gtest_Debug_x86.bat	Run 7
8	Run Tests: Release x86	%MLIB%\resource\utilities\BatchFiles\RunGtest\Run_gtest_Release_x86.bat	Run 8
9	Generate Release Notes	%MLIB%\resource\utilities\BatchFiles\ReleaseNotes\GenerateReleaseNotes_Everything.bat	Run 9
10	LONDON Deploy Excel Add-In	%MLIB%\resource\utilities\BatchFiles\DeployAddin\LONDON_DeployEtradingAddin_UATandPROD.bat	Run 10
11	TOKYO Deploy Excel Add-In	%MLIB%\resource\utilities\BatchFiles\DeployAddin\TOKYO_DeployEtradingAddin_UATandPROD.bat	Run 11
12	Godzilla'sMLIB: Renew Licence	%MLIB%\resource\utilities\BatchFiles\Godzilla'sMLIB\RenewExcelLicenceEtradingAddin.bat	Run 12

Test Harness

- Parallelized Tests, Debug Support, Test Reset Features
- Test Failure Log & Explain



MLIB Ribbon



Features

- [Swap Pricing](#) - Open the Swap Pricer
- [MLIB Browser](#) - Navigate to the MLIB Examples Folder
- [Swap Pricing](#) - Open the Swap Pricer
- [Clear MLIB Cache](#) - Clear the MLIB Object Memory
- [Sticky Handles](#) - Toggle Object Safety Feature On/Off
- [MLIB Version](#) - Display the MLIB Version and License Expiry Date!
- [Record MLIB Functions](#) - Record / Log MLIB Functions

MLIB Functions (not an exhaustive list)

meDate...

meDateBusinessDays
meDateCentralBank
meDateECB
meDateFromTenor
meDateFromYearFraction
meDateFuturesContract
meDateIMMFromMonth
meDateIsHoliday
meDateIsWeekday
meDateIsWeekend
meDateIsWorkingDay
meDateNthIMM
meDateNthIMMFromStartDate
meDateYearFraction

meUtilities...

meUtilityClearLWOCache
meUtilityLVB
meUtilityResize
meUtilityRecord
meUtilityReplay
meUtilityVersion

meCurve...

meCurveCalibrateOIS
meCurveCalibrateSwap
meCurveCalibrateBasis
meCurveCalibrateFXForwards
meCurveDiscountFactors
meCurveForwardRates

meLWO...

meLWOExists
meLWOType
meLWOList
meLWODelete
meLWODeleteAll
meLWOSave
meLWOLoad

meLWOSchedule

meLWOScheduleCreate
meLWOScheduleCreateBespoke
meLWOScheduleDisplay

meLWOCurve...

meLWOCurveCreateOIS
meLWOCurveCreateSwap
meLWOCurveCreateBasis
meLWOCurveCreateFXForwards
meLWOCurveDiscountFactors
meLWOCurveForwardRates
meLWOCurveCalibrate
meLWOCurveMarketDataCreate
meLWOCurveMarketDataDisplay
meLWOCurveDisplay
meLWOCurveGeneratorCreate
meLWOCurveGeneratorDisplay

Functions groups used for navigational ease

meLWOSwap...

meLWOSwapCreate
meLWOSwapCreateFromGenerator
meLWOSwapAddFee
meLWOSwapDisplayCashflows
meLWOSwapDisplay
meLWOSwapAnnuity
meLWOSwapParRate
meLWOSwapParSpread
meLWOSwapPV01
meLWOSwapDV01
meLWOSwapDeltaLadder

meLWOBond...

meLWOBondCreate
meLWOBondDirtyPrice
meLWOBondCleanPrice
meLWOBondAccruedInterest
meLWOBondAccruedInterestDays
meLWOBondDisplaySchedule
meLWOBondDisplayCashflows
meLWOBondYield
meLWOBondCompoundYield
meLWOBondModifiedDuration
meLWOBondDV01
meLWOBondDV01Numerical

meLWOFixingTable

meLWOFixingTableCreate
meLWOFixingTableDisplay
meLWOFixingTableValues

meLWOXccy

meLWOXccySwapCreate
meLWOXccySwapParSpread
meLWOXccySwapSpread

meLWOFee

meLWOFeeLegCreate
meLWOFeeScheduleCreate

Holiday Data Sourced from SwapsWire

☐ ADB (Adelaide)	☐ HKB (Hong Kong)	☐ PAB (Paris)
☐ AEB (AbuDhabi)	☐ HKS (Hong Kong Stock Exchange trading)	☐ PEB (Perth)
☐ AMB (Amsterdam)	☐ HAB (Hanoi)	☐ PRB (Prague)
☐ ANB (Ankara)	☐ HEB (Helsinki)	☐ RJB (Rio de Janeiro)
☐ ATB (Athens)	☐ HMB (Hamilton(Bermuda))	☐ RIB (Riyadh)
☐ AUB (Auckland)	☐ HOB (Hobart)	☐ RKB (Reykjavik)
☐ BAB (Beunos Aires)	☐ ISB (Istanbul)	☐ SOM (Eurex trading (German equity contracts))
☐ BMA (US Treasuries (SIFMA recommended close))	☐ JAB (Jakarta)	☐ SON (Eurex trading (STOXX contracts))
☐ BBB (Brisbane)	☐ JOB (Johannesburg)	☐ SPB (Sao Paulo)
☐ BCB (Bucharest)	☐ KLB (Kuala Lumpur)	☐ SAB (Santiago)
☐ BDB (Budapest)	☐ LAB (Labuan)	☐ SEB (Seoul)
☐ BEB (Beijing)	☐ LBB (Lubljana)	☐ SHB (Shanghai)
☐ BKB (Bangkok)	☐ LMB (Lima)	☐ SIB (Singapore)
☐ BMB (Mumbai)	☐ LNB (London)	☐ SOB (Sofia)
☐ BOB (Bogota)	☐ LSB (Lisbon)	☐ STB (Stockholm)
☐ BRB (Brussels)	☐ LXB (Luxembourg)	☐ SXB (Stuttgart)
☐ BTB (Bratislava)	☐ MDB (Madrid)	☐ SYB (Sydney)
☐ BXB (Brasilia)	☐ MEB (Melbourne)	☐ TYO (ISDA Standard TYO)
☐ CMF (Chicago Mercantile Exch. trading (Equity))	☐ MIB (Milan)	☐ TGT (TARGET)
☐ CHS (Chicago Stock Exchange trading)	☐ MNB (Manila)	☒ TKB (Tokyo)
☐ CNB (Canberra)	☐ MOB (Montreal)	☐ TPB (Taipei)
☐ COB (Copenhagen)	☐ MWB (Moscow)	☐ TRB (Toronto)
☐ DEL (Delaware)	☐ MXB (Mexico City)	☐ VIB (Vienna)
☐ DBB (Dubai)	☐ NBR (Brazilian payments system)	☐ WAB (Warsaw)
☐ DRB (Darwin)	☐ NYB (New York)	☐ WEB (Wellington)
☐ DUB (Dublin)	☐ NYS (New York Stock Exchange trading)	☐ ZAB (Zagreb)
☐ FRB (Frankfurt)	☐ NDQ (NASDAQ)	☐ ZUB (Zurich)
☐ GEB (Geneva)	☐ OSB (Oslo)	

BusinessDayAdjustment

MOD_FOLLOWING

FOLLOWING

MOD_PRECEDING

PRECEDING

NO_CHANGE

Daycounts

30/360

ACT/360

ACT/365FJ

ACT/365

ACT/365_ISDA

ACT/ACT

ACT/ACT_ISMA

Date Functions

meDate...

[meDateBusinessDays](#)

[meDateCentralBank](#)

[meDateECB](#)

[meDateFromTenor](#)

[meDateFromYearFraction](#)

[meDateFuturesContract](#)

[meDateIMMFromMonth](#)

[meDateIsHoliday](#)

[meDateIsWeekday](#)

[meDateIsWeekend](#)

[meDateIsWorkingDay](#)

[meDateNthIMM](#)

[meDateNthIMMFromStartDate](#)

[meDateYearFraction](#)

meDateBusinessDays

From	Tue, 6-Jun-17
To	Tue, 13-Jun-17
Calendar	TGT
NumberOfBusinessDays	5

meDateFromTenor

StartDate	Tue, 6-Jun-17
Tenor	6M
BusinessDayAdjustment	MOD_FOLLOWING
Calendar	TGT
ResultDate	Wed, 6-Dec-17

meDateFromYearFraction

StartDate	Tue, 6-Jun-17
YearFraction	0.5
DayCount	ACT/365
ResultDate	Wed, 6-Dec-17

meDateFuturesContract

Ticker	Z9
ResultDate	Wed, 18-Dec-19

meDateIMMFromMonth

Year	2017
Month	3
Calendar	NYB
BusinessDayAdjustment	MOD_FOLLOWING
ResultDate	Wed, 15-Mar-17

meDateCentralBank

CentralBankID	ECB
ReferenceDate	Tue, 6-Jun-17
NextCentralBankDate	Thu, 8-Jun-17

meDateECB

ReferenceDate	Tue, 6-Jun-17
NextECBDate	Thu, 8-Jun-17

Swap Schedule Generation

meProductSwapSchedule

EffectiveDate	06-Jun-17
MaturityDate	2Y
AccrualDaycount	ACT/360
AccrualFrequency	Quarterly
AccrualBusinessDayAdjustment	MOD_FOLLOWING
AccrualCalendar	TGT
PaymentFrequency	Quarterly
PaymentBusinessDayAdjustment	MOD_FOLLOWING
PaymentCalendar	TGT
PaymentLag	0D
FixingBusinessDayAdjustment	MOD_FOLLOWING
FixingCalendar	TGT
FixingLag	0D
FixingAdvanceOrArrear	ADVANCE
FirstStubDate	
LastStubDate	
RollDay	IMM

FixingDate	AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate
Tue, 6-Jun-17	Tue, 6-Jun-17	Wed, 21-Jun-17	15	0.0417	Wed, 21-Jun-17
Wed, 21-Jun-17	Wed, 21-Jun-17	Wed, 20-Sep-17	91	0.2528	Wed, 20-Sep-17
Wed, 20-Sep-17	Wed, 20-Sep-17	Wed, 20-Dec-17	91	0.2528	Wed, 20-Dec-17
Wed, 20-Dec-17	Wed, 20-Dec-17	Wed, 21-Mar-18	91	0.2528	Wed, 21-Mar-18
Wed, 21-Mar-18	Wed, 21-Mar-18	Wed, 20-Jun-18	91	0.2528	Wed, 20-Jun-18
Wed, 20-Jun-18	Wed, 20-Jun-18	Wed, 19-Sep-18	91	0.2528	Wed, 19-Sep-18
Wed, 19-Sep-18	Wed, 19-Sep-18	Wed, 19-Dec-18	91	0.2528	Wed, 19-Dec-18
Wed, 19-Dec-18	Wed, 19-Dec-18	Wed, 20-Mar-19	91	0.2528	Wed, 20-Mar-19
Wed, 20-Mar-19	Wed, 20-Mar-19	Thu, 6-Jun-19	78	0.2167	Thu, 6-Jun-19

Featuring

- Version and License Expiry Information
- Recording, Replay & Logging
- Resizing Data Blocks
- Appending & Gluing Data

meUtilities...

meUtilityClearLWOCache

meUtilityLVB

meUtilityResize

meUtilityRecord

meUtilityReplay

meUtilityVersion

meUtilityClearLWOCache

```
1=meUtilityClearLWOCache()
```

meUtilityVersion

```
MLIB Addin: Version Number: 20170602.2204. Last built on Jun 2 2017 at 22:04:47. License will expire on Oct 1 2017
```

meUtilityRecord

```
meUtilityRecord( Enable=True\False, Folder="C:\Temp\")
```

- Data Table Resizing
- Removes unwanted #N/As

Input

Data	Value1	Value2	#N/A
1	100	10000	#N/A
2	200	40000	#N/A
3	300	90000	#N/A
4	400	160000	#N/A
5	500	250000	#N/A
6	600	360000	#N/A
7	700	490000	#N/A
8	800	640000	#N/A
9	900	810000	#N/A
10	1000	1000000	#N/A
#N/A	#N/A	#N/A	#N/A
#N/A	#N/A	#N/A	#N/A
#N/A	#N/A	#N/A	#N/A

meUtilityResize

Data	Value1	Value2
1	100	10000
2	200	40000
3	300	90000
4	400	160000
5	500	250000
6	600	360000
7	700	490000
8	800	640000
9	900	810000
10	1000	1000000

Label Value Blocks, LVBs

- Within MLIB Data Blocks are called Label Value Blocks
- Data Blocks can be appended or glued together

Input

Label1	Label2	Label3	Label4	Label5
Value1	Value2	Value3	Value4	Value5

meUtilityLVB

Label1	Value1
Label2	Value2
Label3	Value3
Label4	Value4
Label5	Value5

Input1

Label1	Value1
Label2	Value2

Input2

Label3	Value3
Label4	Value4
Label5	Value5

meUtilityLVBFromMultipleKeysValues

Label1	Value1
Label2	Value2
Label3	Value3
Label4	Value4
Label5	Value5

- MLIB supports Multi-Curve Construction with CSA Support
- Yield Curves were discussed at length in our previous Workshop
- See X:/Secure/DIR_MLib_Analytics/Workshop/Workshops_2016-10
- For CSA Calculations we assume FX Forward Invariance

FX Forward Invariance Example for EUR_JPYCSA

$$FwdFX(t, T)^{EUR/JPY} = S \underbrace{\frac{P(t, T)^{EUR_USDCSA}}{P(t, T)^{JPY_USDCSA}}}_{\text{From Xccy Curve}} = S \underbrace{\frac{P(t, T)^{EUR_JPYCSA}}{P(t, T)^{JPY_JPYCSA}}}_{\text{JPY_JPYCSA from OIS Curve}}$$

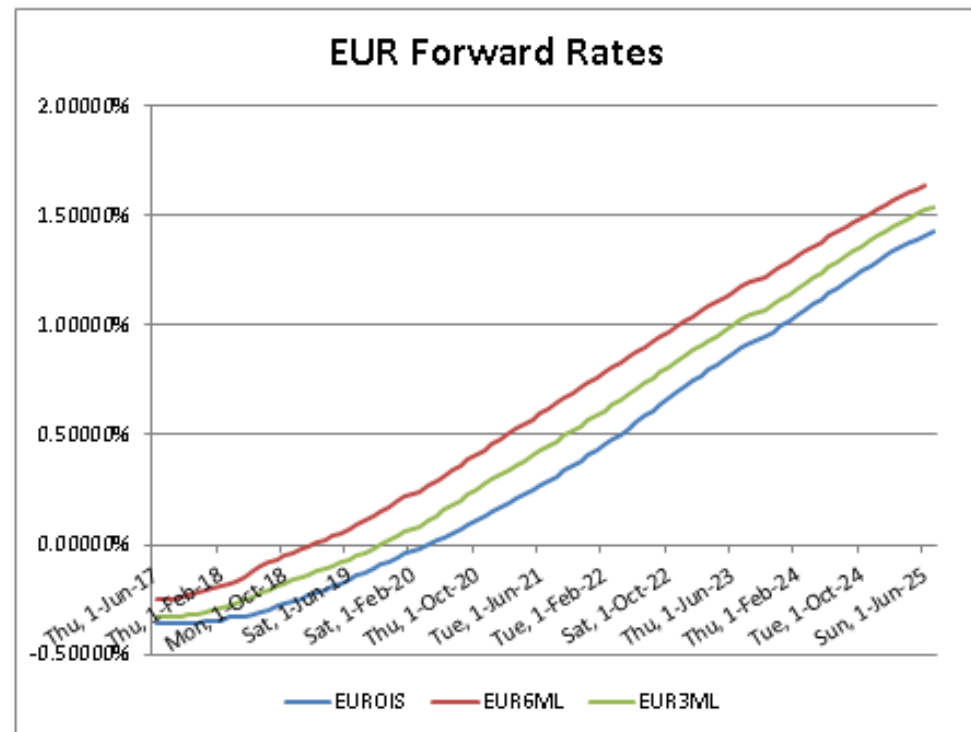
where t denotes the valuation date, S the FX spot rate, $P(t, T)$ the discount factor at time t for tenor T with $0 \leq t \leq T$

- Libor Forecast Rates do not depend on CSA Collateral

Curve Collection **EURYC**

[meCurveForwardRates...](#)

FixingDate	EUROIS	EUR3ML	EUR6ML
Tue, 6-Jun-17	-0.36053%	-0.32903%	-0.25412%
Thu, 6-Jul-17	-0.35574%	-0.32870%	-0.25159%
Mon, 7-Aug-17	-0.36402%	-0.32816%	-0.24522%
Thu, 7-Sep-17	-0.35784%	-0.32664%	-0.23919%
Mon, 9-Oct-17	-0.35989%	-0.32312%	-0.23200%
Thu, 9-Nov-17	-0.35809%	-0.31789%	-0.22169%
Mon, 11-Dec-17	-0.34776%	-0.31091%	-0.21318%
Thu, 11-Jan-18	-0.34745%	-0.30218%	-0.20285%
Mon, 12-Feb-18	-0.35151%	-0.29154%	-0.18924%
Mon, 12-Mar-18	-0.32933%	-0.27988%	-0.17740%
Thu, 12-Apr-18	-0.32995%	-0.26635%	-0.16461%
Mon, 14-May-18	-0.32921%	-0.25084%	-0.14591%
Thu, 14-Jun-18	-0.31889%	-0.23573%	-0.12317%
Mon, 16-Jul-18	-0.30761%	-0.22031%	-0.10188%
Thu, 16-Aug-18	-0.29631%	-0.20561%	-0.08378%
Mon, 17-Sep-18	-0.28426%	-0.19091%	-0.06706%
Wed, 17-Oct-18	-0.27262%	-0.17689%	-0.05261%
Mon, 19-Nov-18	-0.25941%	-0.16198%	-0.03743%
Wed, 19-Dec-18	-0.24704%	-0.14909%	-0.02370%
Mon, 21-Jan-19	-0.23294%	-0.13466%	-0.00820%
Thu, 21-Feb-19	-0.21920%	-0.12156%	0.00683%

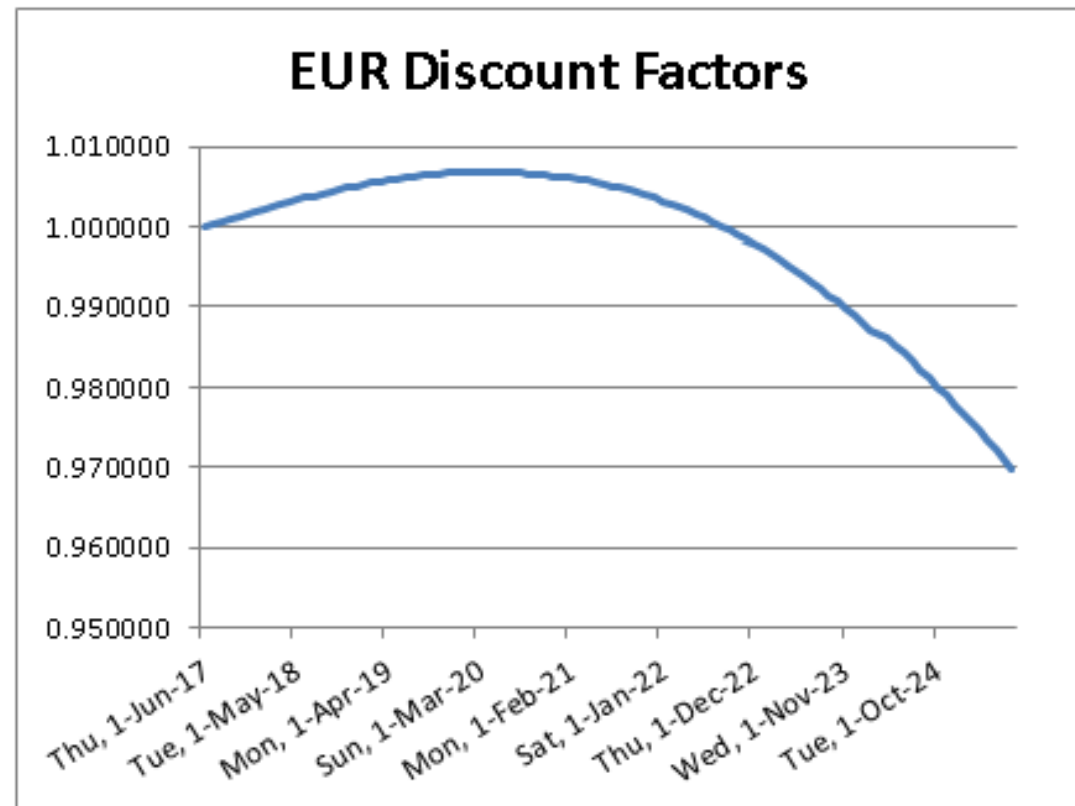


- Discount Factors depend on CSA Collateral

Curve Collection **EURYC**

meCurveDiscountFactors...

Payment Date	EUROIS
Tue, 6-Jun-17	1.000000
Mon, 3-Jul-17	1.000271
Thu, 3-Aug-17	1.000582
Mon, 4-Sep-17	1.000903
Wed, 4-Oct-17	1.001202
Mon, 6-Nov-17	1.001532
Wed, 6-Dec-17	1.001828
Mon, 8-Jan-18	1.002147
Thu, 8-Feb-18	1.002448
Thu, 8-Mar-18	1.002718
Mon, 9-Apr-18	1.003009
Wed, 9-May-18	1.003286
Mon, 11-Jun-18	1.003586
Wed, 11-Jul-18	1.003849
Mon, 13-Aug-18	1.004128
Thu, 13-Sep-18	1.004381



Swap Definition

LegType	Leg1:Fixed	Leg2:Float
Notional	1,000,000	1,000,000
NotionalExchange	NONE	NONE
PayReceive	PAYER	RECEIVER
ForecastCurve	EUR6ML	EUR6ML
DiscountCurve	EUROIS	EUROIS
Currency	EUR	EUR
Leverage	1.00	1.00
FixedRate	1.00%	
FirstFixing		
LastFixing		
FloatSpread		0.00
EffectiveDate	Tue 6-Jun-17	Tue 6-Jun-17
MaturityDate	5Y	5Y
Frequency	Annual	Semi-Annual
Daycount	30/360	ACT/360
FixingBusinessDa		MOD_FOLLOWING
FixingCalendar		TGT
AccrualBusinessD	MOD_FOLLOWING	MOD_FOLLOWING
AccrualCalendar	TGT	TGT
PaymentBusiness	MOD_FOLLOWING	MOD_FOLLOWING

Swap Pricing & Risk

meLWOSwapCreate	Swap Name	IRS_EUR6ML_5Y
	Create Swap	IRS_EUR6ML_5Y@2:0
meLWOSwapPV meLWOSwapPV01 meLWOSwapDV01 meLWOSwapParRate meLWOSwapParSpread meLWOSwapAnnuity	CurveCollection	EURYC
	PV	-41,559
	PV01	502
	DV01	515
	ParRate	0.17262%
	ParSpread	81.5552
	Annuity	5,022,951

- Cashflow displays give a full break down of the Swap calculation

Swap Cashflows

[meLWOSwapDisplayCashflows](#)

LEG1:FIXED Lt													
AccrualStart	AccrualEnc	AccrualDay	AccrualYearFractions	PaymentD	Notional	NotionalExchange	Leverage	FixedRate	Coupon	DiscountFactor	CouponPV		
06-Jun-17	06-Jun-18	365	1.00	06-Jun-18	-1,000,000	0	1.00	1.0%	-10,000	1.003541	-10,035		
06-Jun-18	06-Jun-19	365	1.00	06-Jun-19	-1,000,000	0	1.00	1.0%	-10,000	1.006092	-10,061		
06-Jun-19	08-Jun-20	368	1.01	08-Jun-20	-1,000,000	0	1.00	1.0%	-10,056	1.006808	-10,124		
08-Jun-20	07-Jun-21	364	1.00	07-Jun-21	-1,000,000	0	1.00	1.0%	-9,972	1.005267	-10,025		
07-Jun-21	06-Jun-22	364	1.00	06-Jun-22	-1,000,000	0	1.00	1.0%	-9,972	1.001223	-9,984		
LEG2:FLOAT Lt													
FixingDate	AccrualSta	AccrualEnc	AccrualDays	AccrualYearFractions	PaymentDate	Notional	NotionalExchange	Leverage	FloatSpread	FloatRate	Coupon	DiscountFactor	CouponPV
06-Jun-17	06-Jun-17	06-Dec-17	183	0.5083	06-Dec-17	1,000,000	0	1.00	0.00	-0.25412%	-1,292	1.001828	-1,294
06-Dec-17	06-Dec-17	06-Jun-18	182	0.5056	06-Jun-18	1,000,000	0	1.00	0.00	-0.21453%	-1,085	1.003541	-1,088
06-Jun-18	06-Jun-18	06-Dec-18	183	0.5083	06-Dec-18	1,000,000	0	1.00	0.00	-0.12898%	-656	1.005013	-659
06-Dec-18	06-Dec-18	06-Jun-19	182	0.5056	06-Jun-19	1,000,000	0	1.00	0.00	-0.02968%	-150	1.006092	-151
06-Jun-19	06-Jun-19	06-Dec-19	183	0.5083	06-Dec-19	1,000,000	0	1.00	0.00	0.06209%	316	1.006714	318
06-Dec-19	06-Dec-19	08-Jun-20	185	0.5139	08-Jun-20	1,000,000	0	1.00	0.00	0.17832%	916	1.006808	923
08-Jun-20	08-Jun-20	07-Dec-20	182	0.5056	07-Dec-20	1,000,000	0	1.00	0.00	0.31643%	1,600	1.006330	1,610
07-Dec-20	07-Dec-20	07-Jun-21	182	0.5056	07-Jun-21	1,000,000	0	1.00	0.00	0.45529%	2,302	1.005267	2,314
07-Jun-21	07-Jun-21	06-Dec-21	182	0.5056	06-Dec-21	1,000,000	0	1.00	0.00	0.59260%	2,996	1.003594	3,007
06-Dec-21	06-Dec-21	06-Jun-22	182	0.5056	06-Jun-22	1,000,000	0	1.00	0.00	0.72949%	3,688	1.001223	3,693

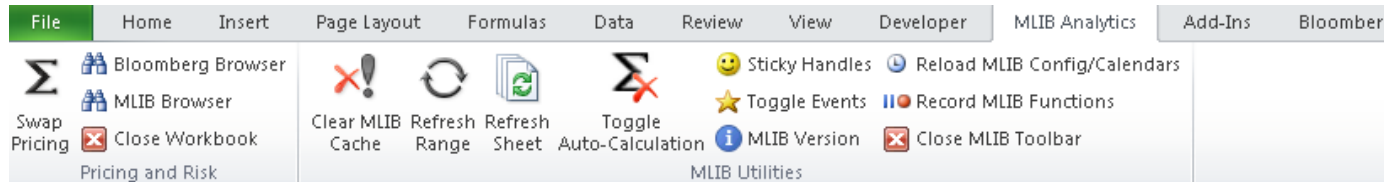
- Bucketed OIS & Libor Delta Risk for a EUR 1mm Eur6mL 3Y IRS

meLWOSwapDeltaLadder

EUR : Pillars	EUR : Total	EUR : Pillars	EUR : Total
EURYC_OIS_OUTRIGHTOISSWAP_ON	0	EURYC_STD_O_N	0
EURYC_OIS_OUTRIGHTOISSWAP_1W	0	EURYC_STD_T_N	0
EURYC_OIS_OUTRIGHTOISSWAP_2W	0	EURYC_STD_LIBOR_1M	0
EURYC_OIS_OUTRIGHTOISSWAP_3W	0	EURYC_STD_LIBOR_3M	0
EURYC_OIS_OUTRIGHTOISSWAP_1M	0	EURYC_STD_LIBOR_6M	4
EURYC_OIS_OUTRIGHTOISSWAP_2M	0	EURYC_STD_LIBOR_12M	0
EURYC_OIS_OUTRIGHTOISSWAP_3M	0	EURYC_STD_FRA_1X7	-5
EURYC_OIS_OUTRIGHTOISSWAP_4M	0	EURYC_STD_FRA_2X8	2
EURYC_OIS_OUTRIGHTOISSWAP_5M	-1	EURYC_STD_FRA_3X9	-1
EURYC_OIS_OUTRIGHTOISSWAP_6M	0	EURYC_STD_FRA_4X10	1
EURYC_OIS_OUTRIGHTOISSWAP_7M	0	EURYC_STD_FRA_5X11	0
EURYC_OIS_OUTRIGHTOISSWAP_8M	0	EURYC_STD_FRA_6X12	1
EURYC_OIS_OUTRIGHTOISSWAP_9M	0	EURYC_STD_FRA_7X13	-2
EURYC_OIS_OUTRIGHTOISSWAP_10M	0	EURYC_STD_FRA_8X14	1
EURYC_OIS_OUTRIGHTOISSWAP_11M	0	EURYC_STD_FRA_9X15	0
EURYC_OIS_OUTRIGHTOISSWAP_12M	10	EURYC_STD_FRA_10X16	1
EURYC_OIS_OUTRIGHTOISSWAP_18M	0	EURYC_STD_FRA_11X17	-1
EURYC_OIS_OUTRIGHTOISSWAP_2Y	21	EURYC_STD_6MLIBORSWAP_3Y	304
EURYC_OIS_OUTRIGHTOISSWAP_3Y	31	EURYC_STD_6MLIBORSWAP_4Y	-2
EURYC_OIS_OUTRIGHTOISSWAP_4Y	0	EURYC_STD_6MLIBORSWAP_5Y	1

- PV01 captures Libor Delta Risk only
- DV01 captures both OIS & Libor Delta Risk

- Accessible from the MLIB Ribbon



- Curve Support: EUR, USD, GBP and JPY LCH

Valuation Date: **Mon, 05-Jun-17**

Bloomberg Access: **TRUE**

Pulse (Seconds): **15**

Calc Mode: **MANUAL**

Use Forwards: **FALSE**
Curve has been set to build using Disco

DATA STREAMING

ENABLE
Data Streaming

CALIBRATION

Step 1
Request Market Data

Step 2
Update Calibration Parameters

Step 3
Build Curves

CURVE BUILDING

Select Currency	Native CSA	CSA Curves
EUR	TRUE	Success
USD	TRUE	Success
GBP	TRUE	Success
JPY	TRUE	Success

Clear Curves & Object Cache

MARKET DATA

OIS CURVE **Ready**

	OIS SWAPS	OIS BASIS	LIBOR SWAPS
EUR	24	17	17
USD	14	12	12
GBP	20	0	0
JPY	20	0	0

SWAP CURVES **Ready**

	MM	FRAX3	FRAX6	FUT	SWAP
EUR	1	0	11	0	17
USD	1	0	0	12	10
GBP	1	0	8	0	16
JPY	1	0	5	0	16

BASIS CURVES **Ready**

	TENOR1	TENOR2	TENOR3	XCCY
EUR	16	14	16	15
USD	15	15	12	-
GBP	17	17	17	15
JPY	17	16	-	16

NAVIGATION

Show / Hide Calibration Instruments

Show / Hide Market Data

Show / Hide Curve Configurations

Show / Hide Curve Build & Charts

Products

- Asset Swaps, OIS, IRS and Tenor Basis, Xccy Swaps (incl. MtM)
- Variable Notional & Bespoke Swaps

SWAP PRICING	ASSET SWAP PRICING	TENOR BASIS SWAPS	VARIABLE NOTIONAL SWAPS	XCCY SWAPS
Show / Hide Swap Pricing Pages	Show / Hide Asset Swap Pricing Pages	Show / Hide Tenor Basis Pricing Pages	Show / Hide VNS Pricing Pages	Show / Hide Xccy Swaps
SWAP CASHFLOWS	ASSET SWAP CASHFLOWS	OVERNIGHT INDEXED SWAPS	STANDARD SWAP GENERATORS	CUSTOMIZED SWAPS
Show / Hide Swap Cashflow Pages	Show / Hide Asset Swap Cashflow Pages	Show / Hide OIS Pricing Pages	Show / Hide Standard Swap Pricing	Show / Hide Customized Swaps

Standard Swaps

- MLIB Supports Swap Templates or 'Generators'
- Simplified Trade Entry for Standard Swaps

Swaps Pricer: IRS Example

EUR Swaps 

PV **-299,129**

EffectiveDate	EffectiveDateOrLag	Notional	Currency	PayReceive	MaturityDate	FixedRate	FloatIndex	FloatSpread	CSA	ParRate	SwapPV	FixedPV	FloatPV	PV01	DV01
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	2Y	1.00000%	6ML	0.00	NOCSA	-0.15920%	23,295	20,096	3,199	201	204
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	3Y	1.00000%	6ML	0.00	NOCSA	-0.06430%	32,134	30,193	1,941	302	308
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	4Y	1.00000%	6ML	0.00	NOCSA	0.04610%	38,366	40,220	-1,854	402	411
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	5Y	1.00000%	6ML	0.00	NOCSA	0.16800%	41,795	50,235	-8,439	502	513
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	6Y	1.00000%	6ML	0.00	NOCSA	0.29360%	42,512	60,181	-17,669	602	613
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	7Y	1.00000%	6ML	0.00	NOCSA	0.42210%	40,470	70,029	-29,559	700	710
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	8Y	1.00000%	6ML	0.00	NOCSA	0.54970%	35,937	79,806	-43,870	798	805
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	9Y	1.00000%	6ML	0.00	NOCSA	0.67180%	29,327	89,357	-60,030	894	895
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	10Y	1.00000%	6ML	0.00	NOCSA	0.78480%	21,250	98,746	-77,496	987	994
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	11Y	1.00000%	6ML	0.00	NOCSA	0.88800%	12,095	107,989	-95,894	1,080	1,096
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	12Y	1.00000%	6ML	0.00	NOCSA	0.98050%	2,283	117,051	-114,769	1,171	1,198
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	15Y	1.00000%	6ML	0.00	NOCSA	1.19080%	-27,317	143,171	-170,488	1,432	1,365
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	20Y	1.00000%	6ML	0.00	NOCSA	1.37150%	-68,136	183,409	-251,545	1,834	1,698
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	25Y	1.00000%	6ML	0.00	NOCSA	1.44080%	-97,026	220,113	-317,139	2,201	2,001
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	30Y	1.00000%	6ML	0.00	NOCSA	1.47080%	-119,498	253,819	-373,317	2,538	2,278
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	40Y	1.00000%	6ML	0.00	NOCSA	1.47580%	-149,374	313,942	-463,316	3,139	2,779
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	50Y	1.00000%	6ML	0.00	NOCSA	1.42880%	-157,242	366,703	-523,945	3,667	3,273

Swaps Pricer: Custom Swap Example

CreateSwap EUR_CUSTOM_SWAP@45:1
DealName EUR_CUSTOM_SWAP
CurveCollection EURYC



Deal Information

	Leg1:Fixed	Leg2:Float
LegType	1,000,000	1,000,000
Notional	1,000,000	1,000,000
NotionalExchange	START_AND_END	START_AND_END
PayReceive	PAY	RECEIVE
EffectiveLag	2D	2D
EffectiveDate	Wed 7-Jun-17	Wed 7-Jun-17
MaturityDate	5Y	5Y
Frequency	Annual	Semi-Annual
ForecastCurve		EUR6ML
DiscountCurve	EUROIS	EUROIS
Currency	EUR	EUR
CompoundMethod	GEOMETRIC	GEOMETRIC
Leverage	1.00	1.00
FixedRate	1.00%	
FirstStubCurveIndex		NATURAL
LastStubCurveIndex		NATURAL
FirstFixing		
LastFixing		
FloatSpread		0.0000
Daycount	30/360	ACT/360
FixingBusinessDayAdjustment		MOD_FOLLOWING
FixingCalendar		TGT
AccrualBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
AccrualCalendar	TGT	TGT
PaymentBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
PaymentCalendar	TGT	TGT
RollDay	NORMAL	NORMAL
FixingLag		0D
PaymentLag	0D	0D
FirstStubDate		
LastStubDate		
StubType	ShortStart	ShortStart

SwapName EUR_CUSTOM_SWAP@45:1
CurveCollection EURYC



Leg1:Fixed

LEG1:FIXED Leg S

AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate	Notional	NotionalExchange	Leverage	FixedRate	Coupon	DiscountFactor	CouponPV
07-Jun-17	07-Jun-18	365	1.0000	07-Jun-17	1,000,000	1.00	1.00	0.0100	1,000,000	1.000018	1,000,018
07-Jun-18	07-Jun-19	365	1.0000	07-Jun-18	-1,000,000	0	1.00	0.0100	-10,000	1.003539	-10,035
07-Jun-19	08-Jun-20	367	1.0028	07-Jun-19	-1,000,000	0	1.00	0.0100	-10,000	1.006070	-10,061
08-Jun-20	07-Jun-21	364	0.9972	08-Jun-20	-1,000,000	0	1.00	0.0100	-10,028	1.006894	-10,097
07-Jun-21	07-Jun-22	365	1.0000	07-Jun-21	-1,000,000	0	1.00	0.0100	-9,972	1.005487	-10,027
				07-Jun-22	-1,000,000	-1,000,000	1.00	0.0100	-1,010,000	1.001486	-1,011,501

Leg2:Float

LEG2:FLOAT Leg:

FixingDate	AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate	Notional	NotionalExchange	Leverage	FloatSpread	FloatRate	Coupon	DiscountFactor	CouponPV
07-Jun-17	07-Jun-17	07-Dec-17	183	0.5083	07-Jun-17	1,000,000	-1,000,000	1.00	0.0000	-0.25600%	-1,000,000	1.000018	-1,000,018
07-Dec-17	07-Dec-17	07-Jun-18	182	0.5056	07-Dec-17	1,000,000	0	1.00	0.0000	-0.20700%	-1,301	1.001844	-1,304
07-Jun-18	07-Jun-18	07-Dec-18	183	0.5083	07-Jun-18	1,000,000	0	1.00	0.0000	-0.13000%	-661	1.005000	-664
07-Dec-18	07-Dec-18	07-Jun-19	182	0.5056	07-Dec-18	1,000,000	0	1.00	0.0000	-0.03563%	-180	1.006070	-181
07-Jun-19	07-Jun-19	09-Dec-19	185	0.5139	07-Jun-19	1,000,000	0	1.00	0.0000	0.06283%	323	1.006705	325
09-Dec-19	09-Dec-19	08-Jun-20	182	0.5056	09-Dec-19	1,000,000	0	1.00	0.0000	0.18326%	926	1.006894	933
08-Jun-20	08-Jun-20	07-Dec-20	182	0.5056	08-Jun-20	1,000,000	0	1.00	0.0000	0.30667%	1,550	1.006504	1,560
07-Dec-20	07-Dec-20	07-Jun-21	182	0.5056	07-Dec-20	1,000,000	0	1.00	0.0000	0.43969%	2,223	1.005487	2,235
07-Jun-21	07-Jun-21	07-Dec-21	183	0.5083	07-Jun-21	1,000,000	0	1.00	0.0000	0.58014%	2,949	1.003807	2,960
07-Dec-21	07-Dec-21	07-Jun-22	182	0.5056	07-Dec-21	1,000,000	1,000,000	1.00	0.0000	0.71598%	1,003,620	1.001486	1,005,111

The End